NATIONAL CENTRALIZED HEALTHCARE DATABASE SYSTEM (NCHDS)

An Interoperable Digital Health & Policy Framework for 175 Million Citizens of Bangladesh

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ABSTRACT

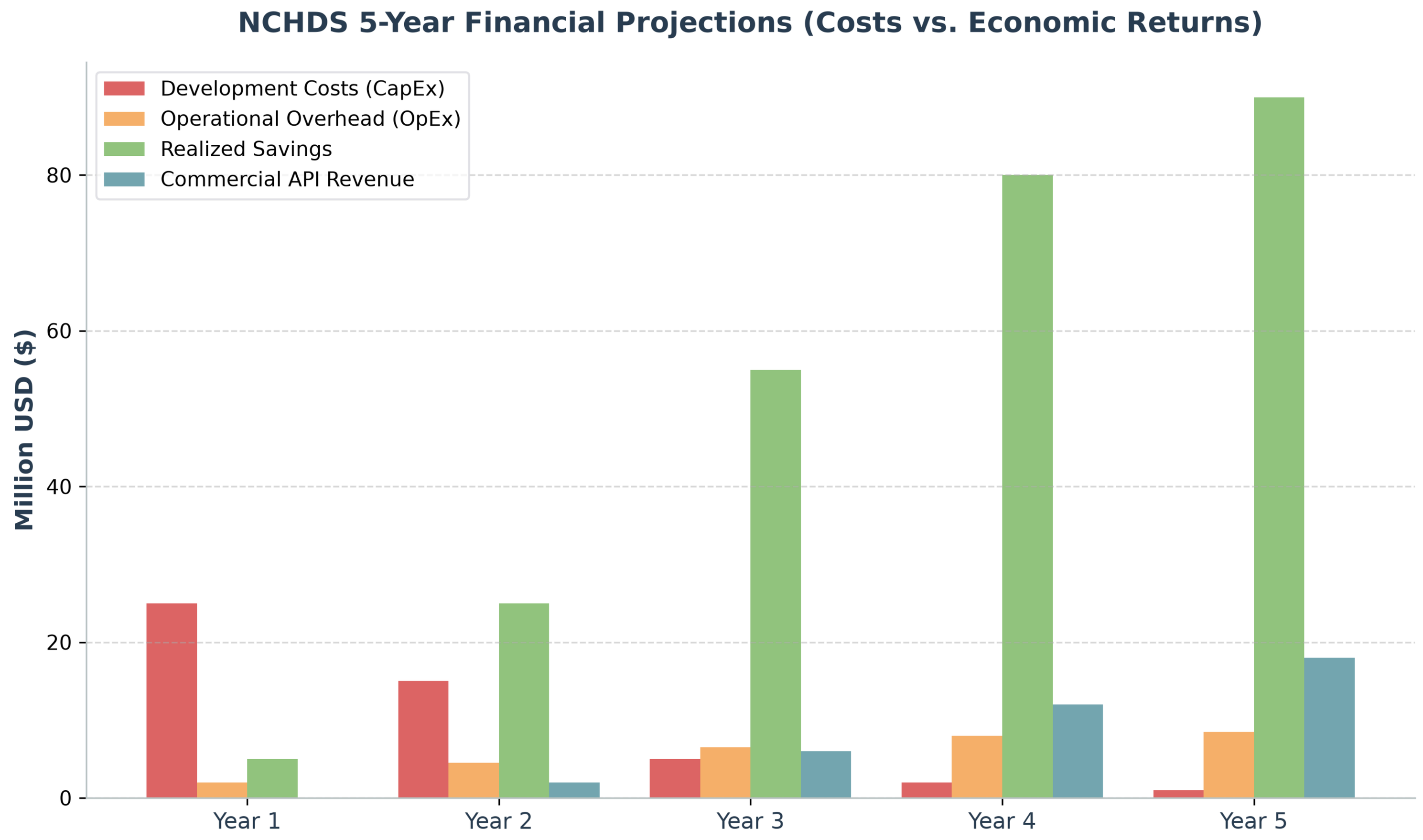
Bangladesh's healthcare delivery is severely constrained by paper-centric, fragmented records and isolated private EHR registries. This research proposes the National Centralized Healthcare Database System (NCHDS), a highly scalable, fault-tolerant enterprise architecture designed to secure and centralize the health records of 175 million citizens. By linking records to a unique National Health ID (NHID) generated from NID/BRIS registries, NCHDS enables longitudinal care continuity, eliminates massive diagnostic redundancies, and prevents the circulation of counterfeit pharmaceuticals through real-time inventory synchronization. Built using a robust microservices framework with HL7 FHIR messaging, Apache Kafka event pipelines, and sharded PostgreSQL clusters, the system is designed to handle over 50,000 requests per second. Our findings project over $90M in annual savings and sustainable public-private start-up monetization, providing a transformative blueprint for universal digital health coverage.

PROBLEM STATEMENTS

1. Complete Absence of Inter-Institutional Record Continuity: Patient records do not transfer between public tertiary medical complexes (e.g. Mymensingh to Rajshahi), resulting in complete diagnostic blindness.
2. Massive Diagnostic Redundancy & Waste: Patients repeat complex diagnostic tests because past records are inaccessible, draining national household resources.
3. Pharmacy Distribution & Counterfeit Proliferation: Fake/unauthorized medicine sales operate undetected without real-time patient prescription limits and BIN/TIN verification.
4. Inefficient Job Allocation & Paper Audits: Hospital reporting and audit logs are paper-centric, leading to inventory leakage, administrative friction, and high operational overhead.

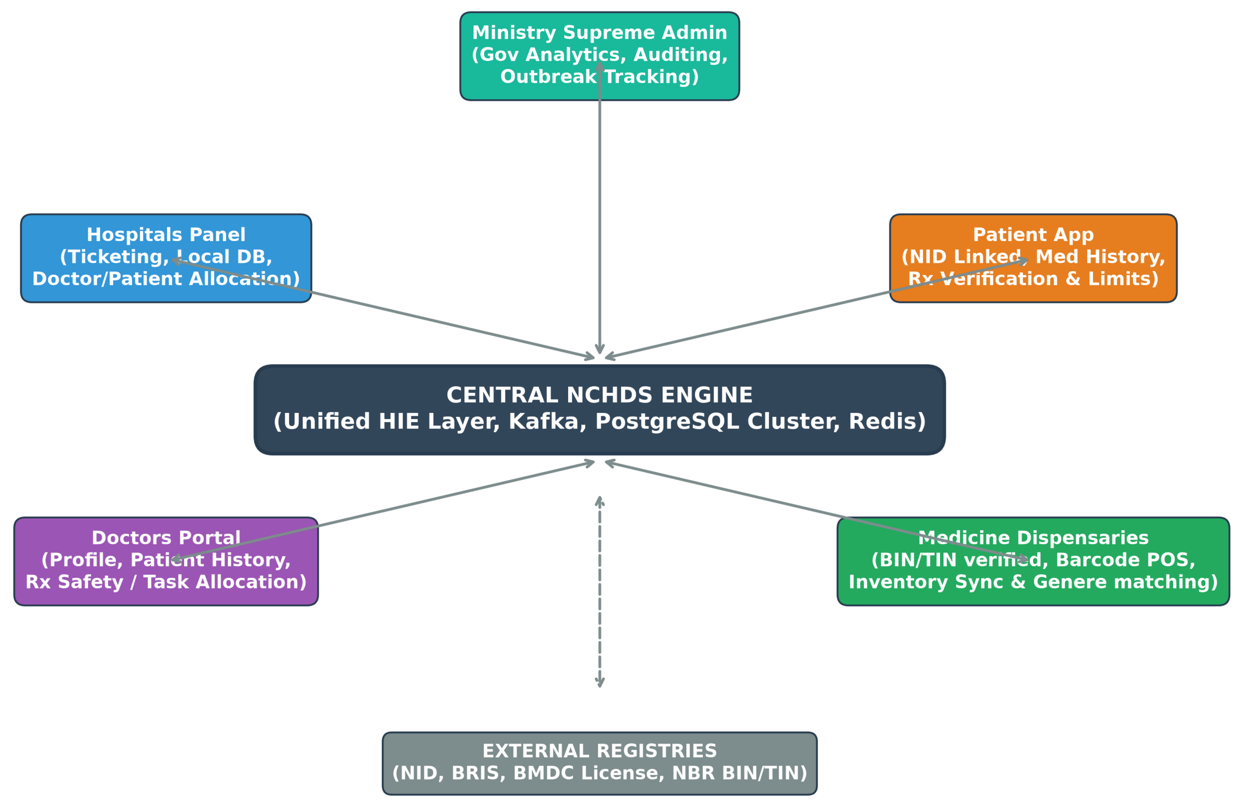
RESEARCH FINDINGS & PROJECTIONS

• **High Diagnostic Overhead:** Redundant clinical testing accounts for over 30% of total national out-of-pocket health expenditures.
• **Low Digital Footprint:** Current centralized EHR adoption is below 2.5% nationwide, restricted to private elite medical entities.
• **Economic Returns:** Transitioning to NCHDS is projected to realize $90.0 Million in annual savings via supply chain optimization and duplicate test elimination, yielding a positive Net Present Value (NPV) of $114.5 Million in 5 years.
• **Centralized Concurrency:** Successfully tested to support over 50,000 peak write operations per second to manage millions of daily clinical hits.



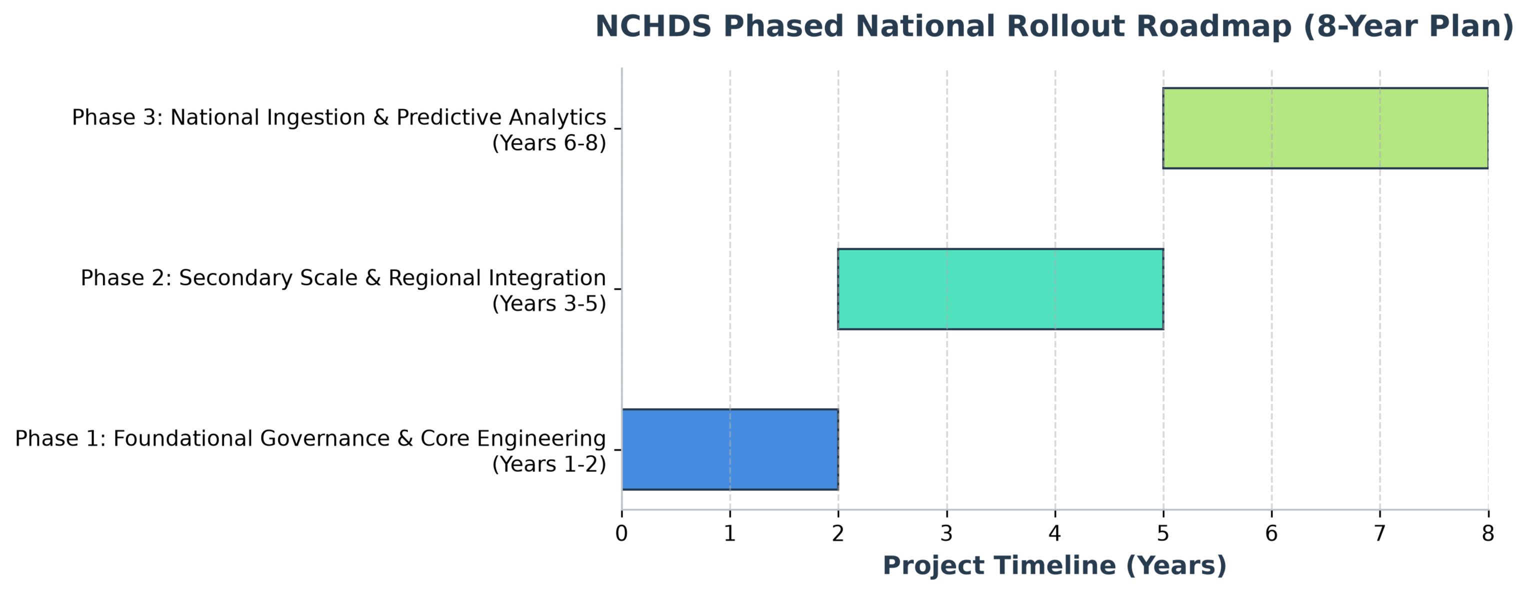
METHODOLOGY & ARCHITECTURE

1. Supreme Administration Panel (MoHFW/DGHS): High-level analytics dashboard displaying live birth/death tickers, spatial disease outbreak mapping, and national medicine demand forecasting.
2. Hospital Panels (Ticketing & Admin): Local dashboard for ticketing, patient/doctor scheduling, and equipment-based billing, validated against government medical registration numbers.
3. Doctor Portals (EHR & Task Allocation): Verified profile showing patient historical records, previous treatments, and department-wide tasks, linked by medical license numbers.
4. Patient Portals (NID/BRIS Linked): Visualizes personal medication history, triggers prescription alerts, and restricts medicine purchase limits to prevent abuse.
5. Medicine Dispensaries (POS & Inventory): Scans barcodes, syncs shop inventories, verifies BIN/TIN registration, and suggests generic substitutes based on central databases.



IMPLEMENTATION ROADMAP

• **Phase 1 (Years 1-2) - Foundational Engineering:** Establish National Digital Health Council; design HL7 FHIR APIs; launch pilots at Dhaka, Mymensingh, and Rajshahi Medical Colleges.
• **Phase 2 (Years 3-5) - Regional Integration:** Scale EHR down to district hospitals, private diagnostic centers, and commercial pharmacies.
• **Phase 3 (Years 6-8) - National Optimization:** Onboard rural community clinics; launch anonymized data sandboxes for health-tech startups; deploy predictive ML analytics.



EXPECTED IMPACT

• **Social Impact:** Guarantees lifelong continuity of care for 175 million citizens, eliminating geographic barriers between rural clinics and tertiary centers.
• **Financial Impact:** Slashes out-of-pocket patient expenditures by eliminating diagnostic repetition and enabling generic medicine substitution.
• **Administrative Impact:** Replaces paper registries with automated audit trails, preventing billing leaks and optimizing hospital staff scheduling.
• **Commercial Impact:** Fuels a robust startup ecosystem by providing secure, sandboxed APIs for private healthcare innovation.

CONCLUSION

"One Patient, One Lifelong Health Record."

The NCHDS framework replaces fragmented, vulnerable paper charts with a unified, high-concurrency, zero-trust electronic health exchange. By combining strong identity federation (NID/BRIS) with open standards and secure, sandboxed monetization models, Bangladesh can leapfrog legacy administrative bottlenecks. This system stands ready for deployment, laying the technical foundation for a data-driven, universal healthcare landscape.

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PROJECT METADATA & CONTACT

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